Problem–10

Title

GCD of Array Elements (Using Euclidean Algorithm)

Topics

GCD, Euclidean Algorithm, Loops, Conditional Statements, Mathematical Reduction

Difficulty Level

Medium

Problem Detailed Statement (Scenario Based)

In a technical aptitude session, students are given a list of integers and are asked to compute the **Greatest Common Divisor (GCD)** of all the numbers.

The **Greatest Common Divisor (GCD)** of two or more integers is the largest positive integer that divides each of the integers without leaving a remainder.

For example:

- GCD of 12 and 18 = 6
- GCD of 24, 36, and 60 = 12
- GCD of 7 and 13 = 1

The trainer emphasizes that students must:

1. Implement the **Euclidean Algorithm** using loops.
2. Avoid using built-in GCD libraries.
3. Compute GCD iteratively:
    - First find GCD of first two numbers.
    - Then compute GCD of result with third number.
    - Continue until all elements are processed.
4. Handle edge cases carefully:
    - If any number is 0.
    - If all numbers are 0.
    - If N = 1.

Euclidean Algorithm logic reminder:

```
while (b != 0)
{
```

```
    temp = b;
    b = a % b;
    a = temp;
}
```

This problem strengthens:

- Mathematical reasoning
- Loop-based algorithm design
- Iterative reduction technique
- Edge-case analysis
- Time complexity understanding

Students must ensure that their program works efficiently even when numbers are large.

Input Format

```
N
a1 a2 a3 ... aN
```

Constraints:

```
1 ≤ N ≤ 1000
0 ≤ ai ≤ 10^9
```

Output Format

- Print a single integer representing the GCD of all given numbers.

Examples

Example 1

Input:

```
3
24 36 60
```

Output:

```
12
```

Example 2

Input:

```
2
7 13
```

Output:

```
1
```

Example 3

Input:

```
4
0 0 0 0
```

Output:

```
0
```

Example 4

Input:

```
1
15
```

Output:

```
15
```